SOM 787 Spatial Analytics-I Class Assignment 01

Objective: Analysing changes of Prevalence of crimes against women across districts and states in India from 2020 to 2022

Install and Load packages

Install all the necessary libraries

```
# Define a vector of required packages
packages <- c(
  "tidyverse",    # For data manipulation and visualization
  "sf",           # For simple feature operations (handling spatial data)
  "here",         # For managing file paths
  "patchwork",    # For organizing multiple plots
  "raster",       # For handling raster data
  "exactextractr",# For extracting data from raster objects
  "tidygeocoder", # For geocoding (converting addresses to geographic coordinates)
  "ggmap",        # For spatial visualization and mapping
  "ggspatial",    # For adding spatial context to ggplot2 visualizations
  "epitrix",      # For epidemiological analysis
  "janitor",      # For data cleaning
  "leaflet",      # For interactive maps
  "HistData",     # For historical datasets
  "gtsummary",     # For creating summary tables
  "spdep",
  "tmap"
)
```

```
# Extract not installed packages
not_installed <- packages[!(packages %in% installed.packages()[ , "Package"])]

# Install not installed packages
if(length(not_installed)) install.packages(not_installed)
```

Load packages

```
suppressPackageStartupMessages(lapply(packages, library, character.only = TRUE))
```

```
## Warning: package 'ggplot2' was built under R version 4.3.3

## Warning: package 'lubridate' was built under R version 4.3.3

## Warning: package 'sf' was built under R version 4.3.3

## Warning: package 'here' was built under R version 4.3.3

## Warning: package 'patchwork' was built under R version 4.3.3

## Warning: package 'sp' was built under R version 4.3.2

## Warning: package 'exactextractr' was built under R version 4.3.3
```

```
## Warning: package 'tidygeocoder' was built under R version 4.3.3

## Warning: package 'ggmap' was built under R version 4.3.3

## Warning: package 'ggspatial' was built under R version 4.3.3

## Warning: package 'epitrix' was built under R version 4.3.3

## Warning: package 'janitor' was built under R version 4.3.3

## Warning: package 'leaflet' was built under R version 4.3.3

## Warning: package 'HistData' was built under R version 4.3.3

## Warning: package 'gtsummary' was built under R version 4.3.3

## Warning: package 'spdep' was built under R version 4.3.3

## Warning: package 'spData' was built under R version 4.3.3

## Warning: package 'tmap' was built under R version 4.3.2

## [[1]]
##  [1] "lubridate" "forcats"   "stringr"   "dplyr"     "purrr"     "readr"
##  [7] "tidyr"     "tibble"    "ggplot2"   "tidyverse" "stats"     "graphics"
## [13] "grDevices" "utils"     "datasets"  "methods"   "base"
##
## [[2]]
##  [1] "sf"        "lubridate" "forcats"   "stringr"   "dplyr"     "purrr"
##  [7] "readr"     "tidyr"     "tibble"    "ggplot2"   "tidyverse" "stats"
## [13] "graphics"  "grDevices" "utils"     "datasets"  "methods"   "base"
##
## [[3]]
##  [1] "here"      "sf"        "lubridate" "forcats"   "stringr"   "dplyr"
##  [7] "purrr"     "readr"     "tidyr"     "tibble"    "ggplot2"   "tidyverse"
## [13] "stats"     "graphics"  "grDevices" "utils"     "datasets"  "methods"
## [19] "base"
##
## [[4]]
##  [1] "patchwork" "here"      "sf"        "lubridate" "forcats"   "stringr"
##  [7] "dplyr"     "purrr"     "readr"     "tidyr"     "tibble"    "ggplot2"
## [13] "tidyverse" "stats"     "graphics"  "grDevices" "utils"     "datasets"
## [19] "methods"   "base"
##
## [[5]]
##  [1] "raster"    "sp"        "patchwork" "here"      "sf"        "lubridate"
##  [7] "forcats"   "stringr"   "dplyr"     "purrr"     "readr"     "tidyr"
## [13] "tibble"    "ggplot2"   "tidyverse" "stats"     "graphics"  "grDevices"
## [19] "utils"     "datasets"  "methods"   "base"
##
## [[6]]
##  [1] "exactextractr" "raster"        "sp"            "patchwork"
##  [5] "here"          "sf"            "lubridate"     "forcats"
##  [9] "stringr"       "dplyr"         "purrr"         "readr"
## [13] "tidyr"         "tibble"        "ggplot2"       "tidyverse"
## [17] "stats"         "graphics"      "grDevices"     "utils"
## [21] "datasets"      "methods"       "base"
##
## [[7]]
```

```
##  [1] "tidygeocoder"  "exactextractr" "raster"        "sp"
##  [5] "patchwork"     "here"          "sf"            "lubridate"
##  [9] "forcats"       "stringr"       "dplyr"         "purrr"
## [13] "readr"         "tidyr"         "tibble"        "ggplot2"
## [17] "tidyverse"     "stats"         "graphics"      "grDevices"
## [21] "utils"         "datasets"      "methods"       "base"
##
## [[8]]
##  [1] "ggmap"         "tidygeocoder"  "exactextractr" "raster"
##  [5] "sp"            "patchwork"     "here"          "sf"
##  [9] "lubridate"     "forcats"       "stringr"       "dplyr"
## [13] "purrr"         "readr"         "tidyr"         "tibble"
## [17] "ggplot2"       "tidyverse"     "stats"         "graphics"
## [21] "grDevices"     "utils"         "datasets"      "methods"
## [25] "base"
##
## [[9]]
##  [1] "ggspatial"     "ggmap"         "tidygeocoder"  "exactextractr"
##  [5] "raster"        "sp"            "patchwork"     "here"
##  [9] "sf"            "lubridate"     "forcats"       "stringr"
## [13] "dplyr"         "purrr"         "readr"         "tidyr"
## [17] "tibble"        "ggplot2"       "tidyverse"     "stats"
## [21] "graphics"      "grDevices"     "utils"         "datasets"
## [25] "methods"       "base"
##
## [[10]]
##  [1] "epitrix"       "ggspatial"     "ggmap"         "tidygeocoder"
##  [5] "exactextractr" "raster"        "sp"            "patchwork"
##  [9] "here"          "sf"            "lubridate"     "forcats"
## [13] "stringr"       "dplyr"         "purrr"         "readr"
## [17] "tidyr"         "tibble"        "ggplot2"       "tidyverse"
## [21] "stats"         "graphics"      "grDevices"     "utils"
## [25] "datasets"      "methods"       "base"
##
## [[11]]
##  [1] "janitor"       "epitrix"       "ggspatial"     "ggmap"
##  [5] "tidygeocoder"  "exactextractr" "raster"        "sp"
##  [9] "patchwork"     "here"          "sf"            "lubridate"
## [13] "forcats"       "stringr"       "dplyr"         "purrr"
## [17] "readr"         "tidyr"         "tibble"        "ggplot2"
## [21] "tidyverse"     "stats"         "graphics"      "grDevices"
## [25] "utils"         "datasets"      "methods"       "base"
##
## [[12]]
##  [1] "leaflet"       "janitor"       "epitrix"       "ggspatial"
##  [5] "ggmap"         "tidygeocoder"  "exactextractr" "raster"
##  [9] "sp"            "patchwork"     "here"          "sf"
## [13] "lubridate"     "forcats"       "stringr"       "dplyr"
## [17] "purrr"         "readr"         "tidyr"         "tibble"
## [21] "ggplot2"       "tidyverse"     "stats"         "graphics"
## [25] "grDevices"     "utils"         "datasets"      "methods"
## [29] "base"
##
## [[13]]
```

```
##  [1] "HistData"      "leaflet"       "janitor"       "epitrix"
##  [5] "ggspatial"     "ggmap"         "tidygeocoder"  "exactextractr"
##  [9] "raster"        "sp"            "patchwork"     "here"
## [13] "sf"            "lubridate"     "forcats"       "stringr"
## [17] "dplyr"         "purrr"         "readr"         "tidyr"
## [21] "tibble"        "ggplot2"       "tidyverse"     "stats"
## [25] "graphics"      "grDevices"     "utils"         "datasets"
## [29] "methods"       "base"
##
## [[14]]
##  [1] "gtsummary"     "HistData"      "leaflet"       "janitor"
##  [5] "epitrix"       "ggspatial"     "ggmap"         "tidygeocoder"
##  [9] "exactextractr" "raster"        "sp"            "patchwork"
## [13] "here"          "sf"            "lubridate"     "forcats"
## [17] "stringr"       "dplyr"         "purrr"         "readr"
## [21] "tidyr"         "tibble"        "ggplot2"       "tidyverse"
## [25] "stats"         "graphics"      "grDevices"     "utils"
## [29] "datasets"      "methods"       "base"
##
## [[15]]
##  [1] "spdep"         "spData"        "gtsummary"     "HistData"
##  [5] "leaflet"       "janitor"       "epitrix"       "ggspatial"
##  [9] "ggmap"         "tidygeocoder"  "exactextractr" "raster"
## [13] "sp"            "patchwork"     "here"          "sf"
## [17] "lubridate"     "forcats"       "stringr"       "dplyr"
## [21] "purrr"         "readr"         "tidyr"         "tibble"
## [25] "ggplot2"       "tidyverse"     "stats"         "graphics"
## [29] "grDevices"     "utils"         "datasets"      "methods"
## [33] "base"
##
## [[16]]
##  [1] "tmap"          "spdep"         "spData"        "gtsummary"
##  [5] "HistData"      "leaflet"       "janitor"       "epitrix"
##  [9] "ggspatial"     "ggmap"         "tidygeocoder"  "exactextractr"
## [13] "raster"        "sp"            "patchwork"     "here"
## [17] "sf"            "lubridate"     "forcats"       "stringr"
## [21] "dplyr"         "purrr"         "readr"         "tidyr"
## [25] "tibble"        "ggplot2"       "tidyverse"     "stats"
## [29] "graphics"      "grDevices"     "utils"         "datasets"
## [33] "methods"       "base"
```

Load and check/clean Data

Load Non-Spatial Data

```r
# Load the crime  data
crime_df <- read_csv(here("data", "crime_dateset_india - Sheet1.csv"))
```

```
## Rows: 706 Columns: 10
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (2): DISTRICT, state name
## dbl (8): Sl. No, censuscode, RR_2020, LL_2020, UL_2020, RR_2022, LL_2022, UL...
##
## i Use `spec()` to retrieve the full column specification for this data.
```

```
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
# Explore the crime data
crime_df |>
  glimpse()
```

```
## Rows: 706
## Columns: 10
## $ `Sl. No`     <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17~
## $ DISTRICT     <chr> "Kupwara", "Badgam", "Leh (ladakh)", "Kargil", "Punch", "~
## $ `state name` <chr> "Jammu & Kashmir", "Jammu & Kashmir", "Jammu & Kashmir", ~
## $ censuscode   <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17~
## $ RR_2020      <dbl> 1.13, 1.52, 0.20, 0.20, 0.90, 1.00, 0.69, 1.32, 0.62, 0.8~
## $ LL_2020      <dbl> 1.00, 1.36, 0.09, 0.11, 0.75, 0.86, 0.58, 1.19, 0.48, 0.7~
## $ UL_2020      <dbl> 1.26, 1.68, 0.37, 0.33, 1.06, 1.15, 0.82, 1.45, 0.76, 0.9~
## $ RR_2022      <dbl> 0.78, 1.06, 0.23, 0.25, 0.81, 1.40, 0.44, 0.61, 1.33, 1.0~
## $ LL_2022      <dbl> 0.69, 0.95, 0.12, 0.14, 0.68, 1.25, 0.36, 0.53, 1.15, 0.9~
## $ UL_2022      <dbl> 0.89, 1.19, 0.40, 0.39, 0.95, 1.56, 0.53, 0.69, 1.54, 1.1~
```

```r
# Clean the column names using the 'clean_names' function from the 'janitor' package and display the ne
# This standardizes column names to lower_case_with_underscores
crime_df |>
  janitor::clean_names() |>
  names()
```

```
##  [1] "sl_no"      "district"   "state_name" "censuscode" "rr_2020"
##  [6] "ll_2020"    "ul_2020"    "rr_2022"    "ll_2022"    "ul_2022"
```

```r
# Lowercase all data entries
crime_df <- data.frame(lapply(crime_df, function(x) {
  if (is.character(x)) {
    return(tolower(x))
  } else {
    return(x)
  }
}), stringsAsFactors = FALSE)
```

```r
# Rename columns
colnames(crime_df) <- c("sl_no", "district", "state", "censuscode", "rr_2020", "ll_2020", "ul_2020", "r
```

```r
head(crime_df)
```

```
##   sl_no     district           state censuscode rr_2020 ll_2020 ul_2020 rr_2022
## 1     1      kupwara jammu & kashmir          1    1.13    1.00    1.26    0.78
## 2     2       badgam jammu & kashmir          2    1.52    1.36    1.68    1.06
## 3     3 leh (ladakh) jammu & kashmir          3    0.20    0.09    0.37    0.23
## 4     4       kargil jammu & kashmir          4    0.20    0.11    0.33    0.25
## 5     5        punch jammu & kashmir          5    0.90    0.75    1.06    0.81
## 6     6      rajouri jammu & kashmir          6    1.00    0.86    1.15    1.40
##   ll_2022 ul_2022
## 1    0.69    0.89
## 2    0.95    1.19
## 3    0.12    0.40
## 4    0.14    0.39
## 5    0.68    0.95
## 6    1.25    1.56
```

```r
# Load the India district shape file
india_district_sf <- read_rds(here("spatial_files","india_district_sf.rds"))
```

```r
# Explore the India district shape file
india_district_sf |>
  glimpse()
```

```
## Rows: 726
## Columns: 2
## $ district_unique <chr> "adilabad_telangana", "agar_malwa_madhya_pradesh", "ag~
## $ geometry        <MULTIPOLYGON [°]> MULTIPOLYGON (((78.84098 19..., MULTIPOLY~
```

```r
india_district_sf
```

```
## Simple feature collection with 726 features and 1 field
## Geometry type: GEOMETRY
## Dimension:     XY
## Bounding box:  xmin: 68.09382 ymin: 6.755101 xmax: 97.4115 ymax: 37.07719
## Geodetic CRS:  WGS 84
## # A tibble: 726 x 2
##    district_unique                                                    geometry
##    <chr>                                                    <MULTIPOLYGON [°]>
##  1 adilabad_telangana        (((78.84098 19.7619, 78.85224 19.72753, 78.84944 1~
##  2 agar_malwa_madhya_pradesh (((76.21246 24.2098, 76.21157 24.19905, 76.21654 2~
##  3 agra_uttar_pradesh        (((78.19691 27.40256, 78.21589 27.40048, 78.22315 ~
##  4 ahmedabad_gujarat         (((72.33654 23.10695, 72.35633 23.08818, 72.36506 ~
##  5 ahmednagar_maharashtra    (((74.67067 19.94678, 74.6719 19.93238, 74.67653 1~
##  6 aizawl_mizoram            (((93.2045 24.04956, 93.19152 24.01485, 93.16826 2~
##  7 ajmer_rajasthan           (((74.97485 26.89097, 74.96499 26.87514, 74.95382 ~
##  8 akola_maharashtra         (((77.61086 20.71787, 77.59075 20.7268, 77.5777 20~
##  9 alappuzha_kerala          (((76.37292 9.836902, 76.37764 9.832135, 76.38157 ~
## 10 aligarh_uttar_pradesh     (((78.47747 28.1131, 78.49717 28.10109, 78.51055 2~
## # i 716 more rows
```

```r
# Convert the spatial data into Geodetic CRS
india_district_sf <- india_district_sf |>
  st_transform(4326)

# Check the CRS of the spatial data
india_district_sf |> st_crs()
```

```
## Coordinate Reference System:
##   User input: EPSG:4326
##   wkt:
## GEOGCRS["WGS 84",
##     ENSEMBLE["World Geodetic System 1984 ensemble",
##         MEMBER["World Geodetic System 1984 (Transit)"],
##         MEMBER["World Geodetic System 1984 (G730)"],
##         MEMBER["World Geodetic System 1984 (G873)"],
##         MEMBER["World Geodetic System 1984 (G1150)"],
##         MEMBER["World Geodetic System 1984 (G1674)"],
##         MEMBER["World Geodetic System 1984 (G1762)"],
##         MEMBER["World Geodetic System 1984 (G2139)"],
```

```
##         ELLIPSOID["WGS 84",6378137,298.257223563,
##             LENGTHUNIT["metre",1]],
##         ENSEMBLEACCURACY[2.0]],
##     PRIMEM["Greenwich",0,
##         ANGLEUNIT["degree",0.0174532925199433]],
##     CS[ellipsoidal,2],
##         AXIS["geodetic latitude (Lat)",north,
##             ORDER[1],
##             ANGLEUNIT["degree",0.0174532925199433]],
##         AXIS["geodetic longitude (Lon)",east,
##             ORDER[2],
##             ANGLEUNIT["degree",0.0174532925199433]],
##     USAGE[
##         SCOPE["Horizontal component of 3D system."],
##         AREA["World."],
##         BBOX[-90,-180,90,180]],
##     ID["EPSG",4326]]
```

```r
# To clean the state district column
# Define a list of state and union territories names

state_names <- c(
  # States
  "andhra_pradesh", "arunachal_pradesh", "assam", "bihar", "chhattisgarh", "goa",
  "gujarat", "haryana", "himachal_pradesh", "jharkhand", "karnataka", "kerala",
  "madhya_pradesh", "maharashtra", "manipur", "meghalaya", "mizoram", "nagaland",
  "odisha", "punjab", "rajasthan", "sikkim", "tamil_nadu", "telangana",
  "tripura", "uttar_pradesh", "uttarakhand", "west_bengal",

  # Union Territories
  "andaman_nicobar_islands", "chandigarh", "dadra_nagar_haveli_and_daman_diu",
  "lakshadweep", "nct_delhi", "puducherry", "ladakh", "jammu_kashmir"
)
```

```r
# Function to separate district and state
split_district_state <- function(df, column_name, state_list) {
  df <- df |>
    mutate(
      # Identify the state part in each entry
      state = sapply(str_split(!!sym(column_name), "_"), function(parts) {
        state_match <- ""
        for (i in seq_along(parts)) {
          candidate <- paste(parts[i:length(parts)], collapse = "_")
          if (candidate %in% state_list) {
            state_match <- candidate
            break
          }
        }
        state_match
      }),
      # Remove the state part from the district name
      district = sapply(str_split(!!sym(column_name), "_"), function(parts) {
        state_match <- ""
        for (i in seq_along(parts)) {
          candidate <- paste(parts[i:length(parts)], collapse = "_")
```

```r
          if (candidate %in% state_list) {
            state_match <- candidate
            break
          }
        }
        # Keep only the parts before the state name
        district_parts <- parts[1:(which(parts == unlist(str_split(state_match, "_"))[1])-1)]
        paste(district_parts, collapse = "_")
      })
    )
  return(df)
}
```

```r
# Apply the function
india_district_sf <-
  split_district_state(india_district_sf, "district_unique", state_names)
```

```
## Warning: There were 13 warnings in `stopifnot()`.
## The first warning was:
## i In argument: `district = sapply(...)`.
## Caused by warning in `1:(which(parts == unlist(str_split(state_match, "_"))[1]) - 1)`:
## ! numerical expression has 2 elements: only the first used
## i Run `dplyr::last_dplyr_warnings()` to see the 12 remaining warnings.
```

```r
# Drop the extra column - "district_unique"
india_district_sf <- india_district_sf[, c(2,3,4)]
```

Match the state names in spatial and non spatial data set

```r
# Check the state names that are present in the crime data but not in the spatial data
(crime_df$state |> unique()) %in%  crime_df$state
```

```
##  [1] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [16] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [31] TRUE TRUE TRUE TRUE
```

```r
(crime_df$state |> unique())[!(crime_df$state |> unique()) %in%  india_district_sf$state]
```

```
##  [1] "jammu & kashmir"           "himachal pradesh"
##  [3] "nct of delhi"              "uttar pradesh"
##  [5] "arunanchal pradesh"        "west bengal"
##  [7] "madhya pradesh"            "daman & diu"
##  [9] "dadara & nagar havelli"    "andhra pradesh"
## [11] "tamil nadu"                "andaman & nicobar island"
## [13] NA
```

```r
# Clean the State names so that they can be matched in both the data sets
crime_df <- crime_df |>
  mutate(
    state = case_when(
      state == "jammu & kashmir"             ~ "jammu_kashmir" ,
      state == "himachal pradesh"            ~ "himachal_pradesh",
      state == "nct of delhi"                ~ "nct_delhi",
      state == "uttar pradesh"               ~ "uttar_pradesh",
      state == "arunanchal pradesh"          ~ "arunachal_pradesh" ,
      state == "west bengal"                 ~ "west_bengal" ,
```

```r
      state == "madhya pradesh"               ~ "madhya_pradesh" ,
      state == "daman & diu"                  ~ "dadra_nagar_haveli_and_daman_diu",
      state == "dadara & nagar havelli"       ~ "dadra_nagar_haveli_and_daman_diu" ,
      state == "andhra pradesh"               ~ "andhra_pradesh",
      state == "tamil nadu"                   ~ "tamil_nadu",
      state == "andaman & nicobar island"     ~ "andaman_nicobar_islands",

      TRUE ~ state
    )
  )
```

```r
# Check the state names that are present in the crime data but not in the spatial data
(crime_df$state |> unique()) %in%  crime_df$state
```

```
##  [1] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [16] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [31] TRUE TRUE TRUE
```

```r
(crime_df$state |> unique())[!(crime_df$state |> unique()) %in%  india_district_sf$state]
```

```
## [1] NA
```

Match the district names in spatial and non spatial data set

```r
# Check the district names that are present in the crime data but not in the spatial data
(crime_df$district |> unique()) %in%  crime_df$district
```

```
##   [1] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
##  [16] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
##  [31] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
##  [46] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
##  [61] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
##  [76] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
##  [91] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [106] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [121] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [136] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [151] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [166] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [181] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [196] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [211] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [226] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [241] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [256] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [271] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [286] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [301] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [316] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [331] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [346] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [361] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [376] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [391] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [406] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
```

```
## [421] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [436] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [451] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [466] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [481] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [496] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [511] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [526] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [541] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [556] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [571] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [586] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [601] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [616] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
```

```r
(crime_df$district |> unique())[!(crime_df$district |> unique()) %in%  india_district_sf$district]
```

```
##   [1] "leh (ladakh)"                   "lahul & spiti"
##   [3] "gurdaspur"                      "shahid bhagat singh nagar"
##   [5] "fatehgarh sahib"                "tarn taran"
##   [7] "sahibzada ajit singh nagar"     "tehri garhwal"
##   [9] "garhwal"                        "udham singh nagar"
##  [11] "hardwar"                        "north west"
##  [13] "north east"                     "new delhi"
##  [15] "south west"                     "sawai madhopur"
##  [17] "jyotiba phule nagar"            "gautam buddha nagar"
##  [19] "mahamaya nagar"                 "rae bareli"
##  [21] "kanpur dehat"                   "kanpur nagar"
##  [23] "allahabad"                      "bara banki"
##  [25] "ambedkar nagar"                 "shrawasti"
##  [27] "siddharth nagar"                "sant kabir nagar"
##  [29] "sant ravi das nagar(bhadohi)"   "kansiram nagar"
##  [31] "pashchim champaran"             "purba champaran"
##  [33] "saran (chhapra)"                "kaimur (bhabua)"
##  [35] "west kameng"                    "east kameng"
##  [37] "papum pare"                     "upper subansiri"
##  [39] "west siang"                     "east siang"
##  [41] "upper siang"                    "lower subansiri"
##  [43] "kurung kumey"                   "dibang valley"
##  [45] "lower dibang valley"            "imphal west"
##  [47] "imphal east"                    "lawangtlai"
##  [49] "west tripura"                   "south tripura"
##  [51] "north tripura"                  "west garo hills"
##  [53] "east garo hills"                "south garo hills"
##  [55] "west khasi hills"               "ri bhoi"
##  [57] "east khasi hills"               "jaintia hills"
##  [59] "marigaon"                       "karbi anglong"
##  [61] "dima hasao"                     "kamrup metropolitan"
##  [63] "darjiling"                      "koch bihar"
##  [65] "uttar dinajpur"                 "dakshin dinajpur"
##  [67] "barddhaman"                     "north 24 parganas"
##  [69] "hugli"                          "haora"
##  [71] "south 24 parganas"              "pashchim medinipur"
##  [73] "purba medinipur"                "purbi singhbhum"
##  [75] "pashchimi singhbhum"            "saraikela-kharsawan"
```

```
##  [77] "bauda"                      "janjgir-champa"
##  [79] "uttar bastar kanker"        "dakshin bastar dantewada"
##  [81] "west nimar"                 "narsimhapur"
##  [83] "east nimar"                 "banas kantha"
##  [85] "sabar kantha"               "ahmadabad"
##  [87] "panch mahals"               "dohad"
##  [89] "the dangs"                  "dadra & nagar haveli"
##  [91] "garhchiroli"                "mumbai suburban"
##  [93] "ahmadnagar"                 "rangareddy"
##  [95] "mahbubnagar"                "warangal"
##  [97] "east godavari"              "west godavari"
##  [99] "sri potti sriramulu nellore" "y.s.r."
## [101] "uttara kannada"             "dakshina kannada"
## [103] "chamrajnagar"               "bangalore rural"
## [105] "north goa"                  "south goa"
## [107] "the nilgiris"               "nagappattinam"
## [109] "virudunagar"                "north & middle andaman"
## [111] "south andaman"              NA
```

```r
# Clean the district names so that they can be matched in both the data sets
crime_df <- crime_df |>
  mutate(
    district = case_when(
      district == "ahmadabad"                ~ "ahmedabad",
      district == "ahmadnagar"               ~ "ahmednagar",
      district == "allahabad"                ~ "prayagraj",
      district == "ambedkar nagar"           ~ "ambedkar_nagar",
      district == "banas kantha"             ~ "banaskantha",
      district == "bangalore rural"          ~ "bangalore_rural",
      district == "bara banki"               ~ "barabanki",
      district == "barddhaman"               ~ "barddhaman", # 2017, this was split in purab barddhaman
      district == "bauda"                    ~ "bharuch",
      district == "chamrajnagar"             ~ "chamarajanagar",
      district == "dadra & nagar haveli"     ~ "dadra",
      district == "dakshin bastar dantewada" ~ "dantewada",
      district == "dakshin dinajpur"         ~ "dakshin_dinajpur",
      district == "dakshina kannada"         ~ "dakshina_kannada",
      district == "darjiling"                ~ "darjeeling",
      district == "dibang valley"            ~ "dibang_valley",
      district == "dima hasao"               ~ "dima_hasao",
      district == "dohad"                    ~ "dahod",
      district == "east garo hills"          ~ "east_garo_hills",
      district == "east godavari"            ~ "east_godavari",
      district == "east kameng"              ~ "east_kameng",
      district == "east khasi hills"         ~ "east_khasi_hills",
      district == "east nimar"               ~ "khandwa_east_nimar",
      district == "east siang"               ~ "east_siang",
      district == "fatehgarh sahib"          ~ "fatehgarh_sahib",
      district == "garhchiroli"              ~ "gadchiroli",
      district == "garhwal"                  ~ "pauri_garhwal",
      district == "gautam buddha nagar"      ~ "gautam_buddha_nagar",
      district == "gurdaspur"                ~ "gurudaspur",
      district == "haora"                    ~ "howrah",
      district == "hardwar"                  ~ "haridwar",
```

```
    district == "hugli"                        ~ "hooghly",
    district == "imphal east"                  ~ "imphal_east",
    district == "imphal west"                  ~ "imphal_west",
    district == "jaintia hills"                ~ "jaintia_hills", # here are two east and west
    district == "janjgir-champa"               ~ "janjgir_champa",
    district == "jyotiba phule nagar"          ~ "jyotiba_phule_nagar",
    district == "kaimur (bhabua)"              ~ "kaimur_bhabua",
    district == "kamrup metropolitan"          ~ "kamrup_metropolitan",
    district == "kanpur dehat"                 ~ "kanpur_dehat",
    district == "kanpur nagar"                 ~ "kanpur_nagar",
    district == "kansiram nagar"               ~ "kanshiram_nagar",
    district == "karbi anglong"                ~ "karbi_anglong",
    district == "koch bihar"                   ~ "koch_bihar",
    district == "kurung kumey"                 ~ "kurung_kumey",
    district == "lahul & spiti"                ~ "lahul_spiti",
    district == "lawangtlai"                   ~ "lawngtlai",
    district == "leh (ladakh)"                 ~ "leh",
    district == "lower dibang valley"          ~ "lower_dibang_valley",
    district == "lower subansiri"              ~ "lower_subansiri",
    district == "mahamaya nagar"               ~ "mahamaya_nagar",
    district == "mahbubnagar"                  ~ "mahabubnagar",
    district == "marigaon"                     ~ "morigaon",
    district == "mumbai suburban"              ~ "mumbai_suburban",
    district == "nagappattinam"                ~ "nagapattinam",
    district == "narsimhapur"                  ~ "narsinghpur",
    district == "new delhi"                    ~ "new_delhi",
    district == "north & middle andaman"       ~ "north_middle",
    district == "north 24 parganas"            ~ "north_twenty_four_parganas",
    district == "north east"                   ~ "north_east",
    district == "north goa"                    ~ "north",
    district == "north tripura"                ~ "north",
    district == "north west"                   ~ "north_west",
    district == "panch mahals"                 ~ "panchmahal",
    district == "papum pare"                   ~ "papum_pare",
    district == "pashchim champaran"           ~ "pashchim_champaran",
    district == "pashchim medinipur"           ~ "paschim_medinipur",
    district == "pashchimi singhbhum"          ~ "pashchimi_singhbhum",
    district == "purba champaran"              ~ "purba_champaran",
    district == "purba medinipur"              ~ "purba_medinipur",
    district == "purbi singhbhum"              ~ "purbi_singhbhum",
    district == "rae bareli"                   ~ "rae_bareli",
    district == "rangareddy"                   ~ "ranga_reddy",
    district == "ri bhoi"                      ~ "ribhoi",
    district == "sabar kantha"                 ~ "sabarkantha",
    district == "sahibzada ajit singh nagar" ~ "sahibzada_ajit_singh_nagar",
    district == "sant kabir nagar"             ~ "sant_kabeer_nagar",
    district == "sant ravi das nagar(bhadohi)" ~ "sant_ravidas_nagar_bhadohi",
    district == "saraikela-kharsawan"          ~ "saraikela_kharsawan",
    district == "saran (chhapra)"              ~ "saran",
    district == "sawai madhopur"               ~ "sawai_madhopur",
    district == "shahid bhagat singh nagar" ~ "shahid_bhagat_singh_nagar",
    district == "shrawasti"                    ~ "shravasti",
    district == "siddharth nagar"              ~ "siddharthnagar",
```

```r
      district == "south 24 parganas"         ~ "south_twenty_four_parganas",
      district == "south andaman"             ~ "south",
      district == "south garo hills"          ~ "south_garo_hills",
      district == "south goa"                 ~ "south",
      district == "south tripura"             ~ "south",
      district == "south west"                ~ "south_west",
      district == "sri potti sriramulu nellore" ~ "sri_potti_sriramulu_nellore",
      district == "tarn taran"                ~ "tarn_taran",
      district == "tehri garhwal"             ~ "tehri_garhwal",
      district == "the dangs"                 ~ "the_dangs",
      district == "the nilgiris"              ~ "the_nilgiris",
      district == "udham singh nagar"         ~ "udam_singh_nagar",
      district == "upper siang"               ~ "upper_siang",
      district == "upper subansiri"           ~ "upper_subansiri",
      district == "uttar bastar kanker"       ~ "uttar_bastar_kanker",
      district == "uttar dinajpur"            ~ "uttar_dinajpur",
      district == "uttara kannada"            ~ "uttara_kannada",
      district == "virudunagar"               ~ "virudhunagar",
      district == "warangal"                  ~ "warangal", # there is warangal_urban to be merged
      district == "west garo hills"           ~ "west_garo_hills",
      district == "west godavari"             ~ "west_godavari",
      district == "west kameng"               ~ "west_kameng",
      district == "west khasi hills"          ~ "west_khasi_hills",
      district == "west nimar"                ~ "khargone_west_nimar",
      district == "west siang"                ~ "west_siang",
      district == "west tripura"              ~ "west",
      district == "y.s.r."                    ~ "y_s_r",
      TRUE ~ district
    )
  )
```

```r
# Check the district names that are present in the crime data but not in the spatial data
(crime_df$district |> unique()) %in%  crime_df$district
```

```
##   [1] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
##  [16] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
##  [31] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
##  [46] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
##  [61] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
##  [76] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
##  [91] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [106] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [121] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [136] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [151] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [166] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [181] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [196] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [211] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [226] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [241] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [256] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [271] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [286] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
```

```
## [301] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [316] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [331] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [346] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [361] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [376] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [391] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [406] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [421] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [436] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [451] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [466] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [481] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [496] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [511] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [526] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [541] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [556] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [571] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [586] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [601] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [616] TRUE TRUE TRUE TRUE
```

```r
(crime_df$district |> unique())[!(crime_df$district |> unique()) %in%  india_district_sf$district]
```

```
## [1] "jaintia_hills" "barddhaman"    "warangal"      NA
```

```r
# Merge the districts in spatial data where the non spatial data refers to only one district (i.e befor
merge_districts <- function(sf_data, districts_to_merge, new_district_name) {
  # Check if all districts to merge exist in the dataset
  if (!all(districts_to_merge %in% sf_data$district)) {
    stop("One or more districts to merge are not present in the dataset.")
  }

  # Merge the geometries of the specified districts
  merged_geom <- sf_data |>
    filter(district %in% districts_to_merge) |>
    summarise(geometry = st_union(geometry), .groups = 'drop')

  # Create a new data frame for the merged district
  merged_district <- sf_data |>
    filter(district %in% districts_to_merge) |>
    select(state) |>
    distinct() |>
    mutate(district = new_district_name)

  # Combine the new merged district with the rest of the data
  result_sf <- rbind(
    sf_data |>
      filter(!district %in% districts_to_merge),  # Keep districts that were not merged
    st_sf(merged_district, geometry = merged_geom$geometry)  # Add the merged district
  )

  return(result_sf)
}
```

```r
# Merge Warangal
merged <- merge_districts(india_district_sf, c("warangal_rural", "warangal_urban"), "warangal")
```

```
## Warning in data.frame(..., check.names = FALSE): row names were found from a
## short variable and have been discarded
```

```r
merged <- merge_districts(merged, c("warangal", "warangal"), "warangal")
```

```r
# Merge Jaintia Hills
merged <- merge_districts(merged, c("east_jaintia_hills", "west_jaintia_hills"), "jaintia_hills")
```

```
## Warning in data.frame(..., check.names = FALSE): row names were found from a
## short variable and have been discarded
```

```r
merged <- merge_districts(merged, c("jaintia_hills", "jaintia_hills"), "jaintia_hills")
```

```r
# Merge Barddhaman
merged <- merge_districts(merged, c("paschim_barddhaman", "purba_barddhaman"), "barddhaman")
```

```
## Warning in data.frame(..., check.names = FALSE): row names were found from a
## short variable and have been discarded
```

```r
merged <- merge_districts(merged, c("barddhaman", "barddhaman"), "barddhaman")
```

```r
glimpse(merged)
```

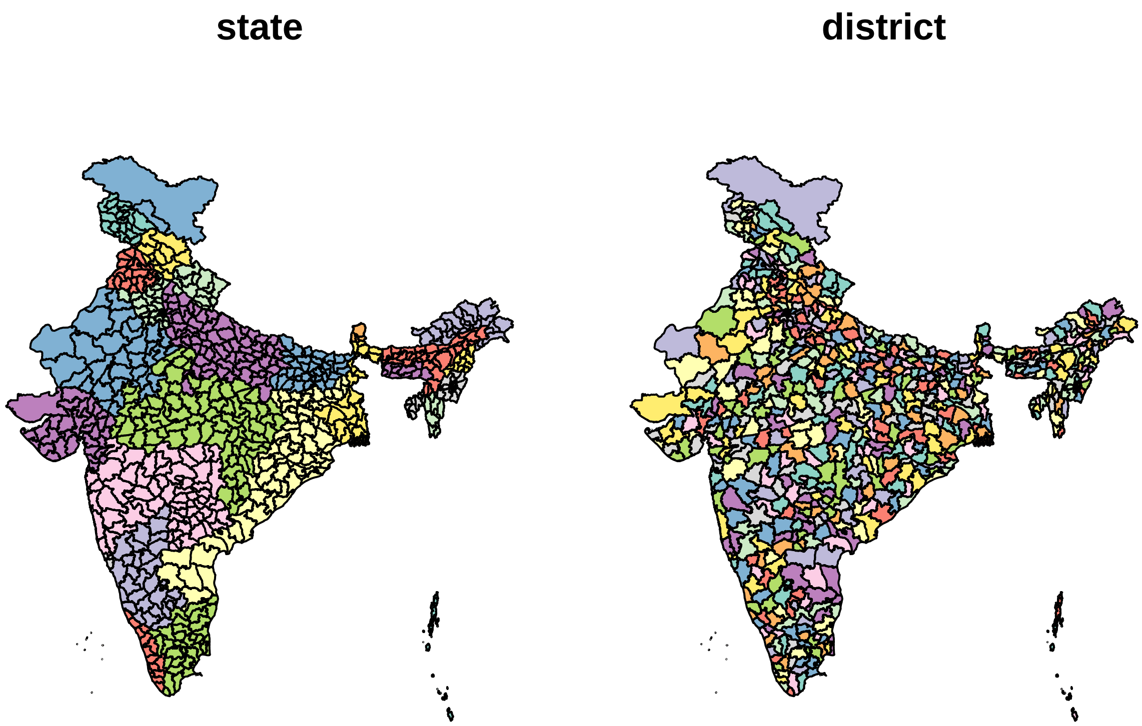
```
## Rows: 723
## Columns: 3
## $ geometry <MULTIPOLYGON [°]> MULTIPOLYGON (((78.84098 19..., MULTIPOLYGON (((~
## $ state    <chr> "telangana", "madhya_pradesh", "uttar_pradesh", "gujarat", "m~
## $ district <chr> "adilabad", "agar_malwa", "agra", "ahmedabad", "ahmednagar", ~
```

```r
# Set the data to MULTIPOLYGON
merged <- st_cast(merged, "MULTIPOLYGON")
```

```r
# Print and plot the result
print(merged)
```

```
## Simple feature collection with 723 features and 2 fields
## Geometry type: MULTIPOLYGON
## Dimension:     XY
## Bounding box:  xmin: 68.09382 ymin: 6.755101 xmax: 97.4115 ymax: 37.07719
## Geodetic CRS:  WGS 84
## # A tibble: 723 x 3
##    state          district                                              geometry
##    <chr>          <chr>                                       <MULTIPOLYGON [°]>
##  1 telangana      adilabad   (((78.84098 19.7619, 78.85224 19.72753, 78.84944 1~
##  2 madhya_pradesh agar_malwa (((76.21246 24.2098, 76.21157 24.19905, 76.21654 2~
##  3 uttar_pradesh  agra       (((78.19691 27.40256, 78.21589 27.40048, 78.22315 ~
##  4 gujarat        ahmedabad  (((72.33654 23.10695, 72.35633 23.08818, 72.36506 ~
##  5 maharashtra    ahmednagar (((74.67067 19.94678, 74.6719 19.93238, 74.67653 1~
##  6 mizoram        aizawl     (((93.2045 24.04956, 93.19152 24.01485, 93.16826 2~
##  7 rajasthan      ajmer      (((74.97485 26.89097, 74.96499 26.87514, 74.95382 ~
##  8 maharashtra    akola      (((77.61086 20.71787, 77.59075 20.7268, 77.5777 20~
##  9 kerala         alappuzha  (((76.37292 9.836902, 76.37764 9.832135, 76.38157 ~
## 10 uttar_pradesh  aligarh    (((78.47747 28.1131, 78.49717 28.10109, 78.51055 2~
## # i 713 more rows
```

```r
plot(merged)
```



Merge the spatial and non-spatial data

```r
# Merge the non-spatial and spatial data based on the 'district' column
district_crime_sf <- merged |>
  st_make_valid() |>
  group_by(district) |>
  summarise() |>
  left_join(crime_df, by = "district")
```

```r
# Set the data to MULTIPOLYGON
district_crime_sf <- st_cast(district_crime_sf, "MULTIPOLYGON")
```

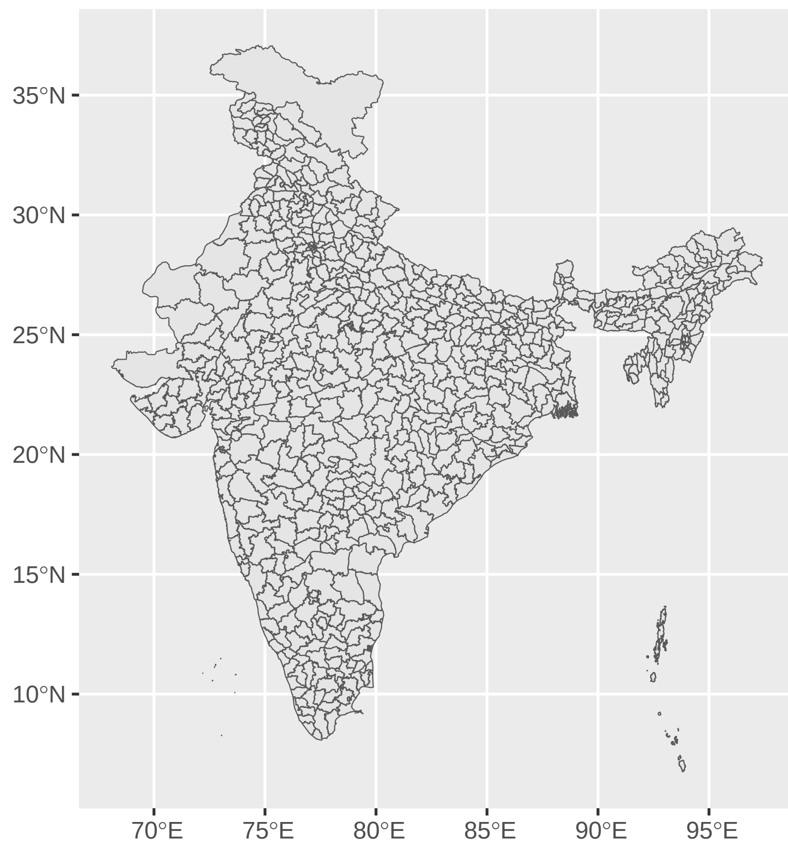
```r
# Drop extra col
district_crime_sf <- district_crime_sf[, -c(2)]
```

```r
# Check the structure of the merged data
district_crime_sf
```

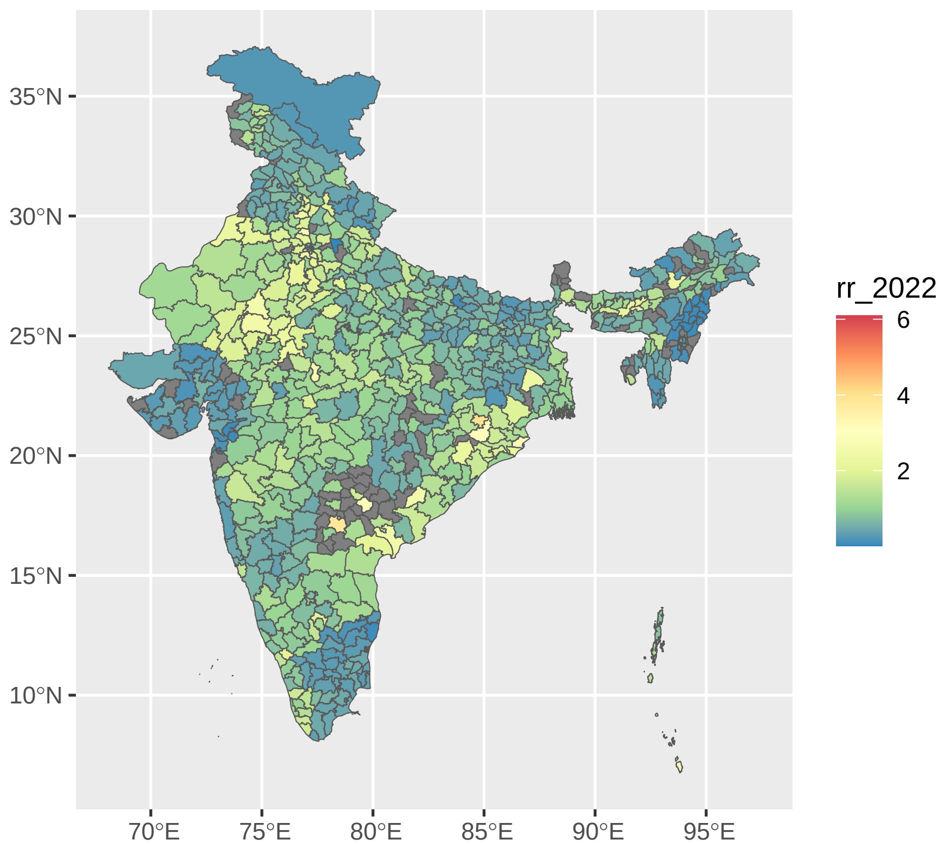
```
## Simple feature collection with 727 features and 9 fields
## Geometry type: MULTIPOLYGON
## Dimension:     XY
## Bounding box:  xmin: 68.09382 ymin: 6.7551 xmax: 97.4115 ymax: 37.07719
## Geodetic CRS:  WGS 84
## # A tibble: 727 x 10
##    district   state   censuscode rr_2020 ll_2020 ul_2020 rr_2022 ll_2022 ul_2022
```

```
##     <chr>      <chr>         <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>
##   1 adilabad   andhra~         532   0.64   0.58   0.69   0.74   0.69   0.8 
##   2 agar_malwa <NA>             NA  NA     NA     NA     NA     NA     NA   
##   3 agra       uttar_~         146   0.77   0.72   0.81   1.01   0.97   1.06
##   4 ahmedabad  gujarat         474   0.77   0.73   0.81   0.62   0.59   0.65
##   5 ahmednagar mahara~         522   1.2    1.15   1.26   1.36   1.3    1.42
##   6 aizawl     mizoram         283   0.52   0.41   0.65   0.44   0.34   0.55
##   7 ajmer      rajast~         119   2.28   2.18   2.39   2.55   2.45   2.65
##   8 akola      mahara~         501   1.32   1.22   1.42   1.18   1.1    1.27
##   9 alappuzha  kerala          598   0.84   0.77   0.91   1.02   0.95   1.1 
##  10 aligarh    uttar_~         143   1.43   1.36   1.5    1.83   1.76   1.91
## # i 717 more rows
## # i 1 more variable: geometry <MULTIPOLYGON [°]>
```

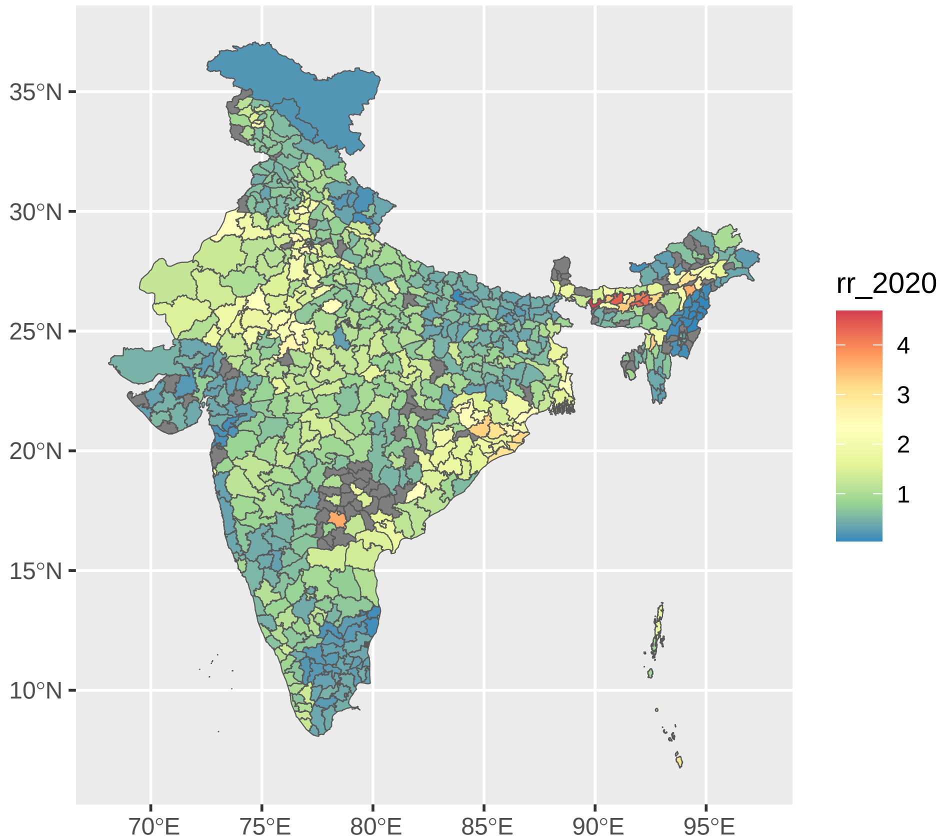
```r
# Plot the map
district_crime_sf |>
  ggplot() +
  geom_sf()
```




```r
# Change color palette for better visualization

# rr_2022 with color palette
district_crime_sf |>
  ggplot() +
  geom_sf(aes(fill = rr_2022)) +
  scale_fill_distiller(palette = "Spectral")
```




```r
# rr_2020 with color palette
district_crime_sf |>
  ggplot() +
  geom_sf(aes(fill = rr_2020)) +
  scale_fill_distiller(palette = "Spectral")
```

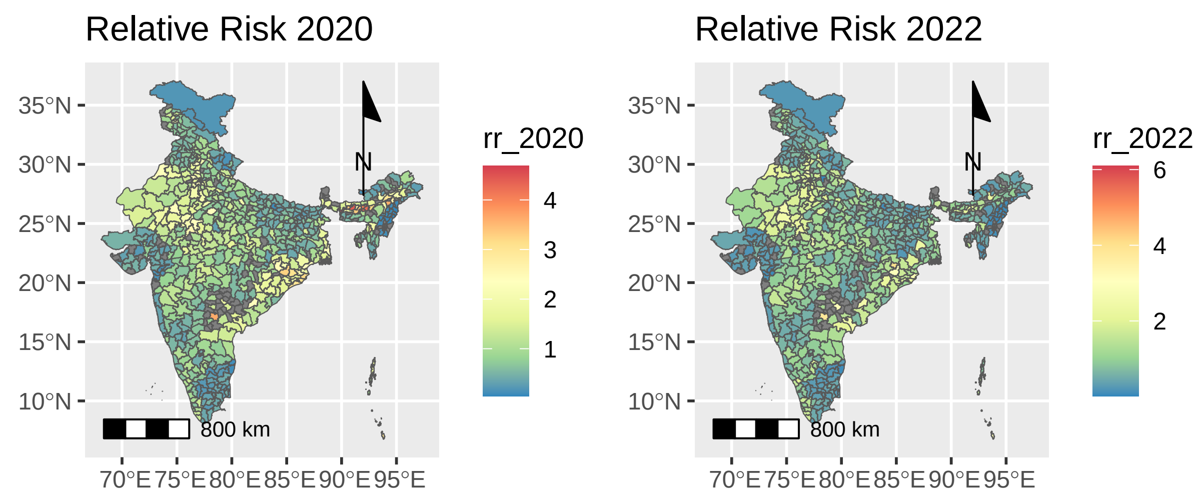



```r
# Plot rr_2022 and rr_2020 side by side using patchwork

# Create individual plots
rr_2022 <- district_crime_sf |>
  ggplot() +
  geom_sf(aes(fill = rr_2022)) +
  scale_fill_distiller(palette = "Spectral") +
  labs(title = "Relative Risk 2022")

rr_2020 <- district_crime_sf |>
  ggplot() +
  geom_sf(aes(fill = rr_2020)) +
  scale_fill_distiller(palette = "Spectral") +
  labs(title = "Relative Risk 2020")


# add North Arrow and Scale from the ggspatial package
rr_2022 <- district_crime_sf |>
  ggplot() +
  geom_sf(aes(fill = rr_2022)) +
  scale_fill_distiller(palette = "Spectral") +
  labs(title = "Relative Risk 2022") +
  annotation_north_arrow(location = "tr", which_north = "true", style = north_arrow_minimal) +
  annotation_scale()

rr_2020 <- district_crime_sf |>
```

```r
  ggplot() +
  geom_sf(aes(fill = rr_2020)) +
  scale_fill_distiller(palette = "Spectral") +
  labs(title = "Relative Risk 2020") +
  annotation_north_arrow(location = "tr", which_north = "true", style = north_arrow_minimal) +
  annotation_scale()
```

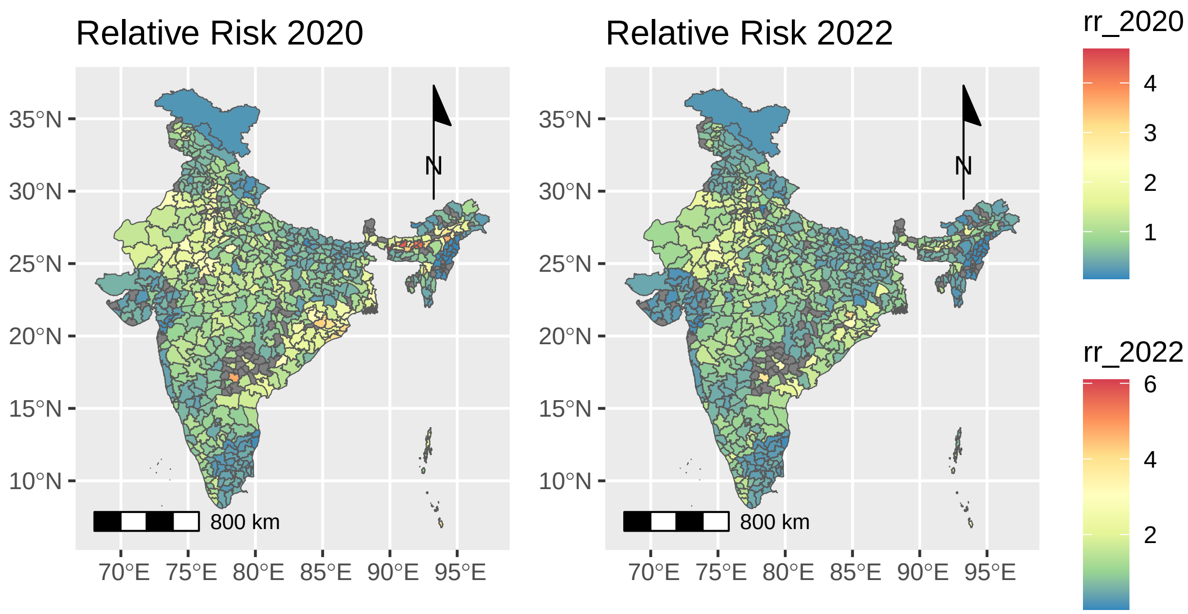
```r
# Combine the plots
rr_2020 +rr_2022
```

```
## Scale on map varies by more than 10%, scale bar may be inaccurate
## Scale on map varies by more than 10%, scale bar may be inaccurate
```




```r
# Adjust scales for a common legend
(rr_2020 + rr_2022) +
  plot_layout(guides = "collect") &
  scale_fill_distiller(palette = "Spectral", limits = range(c(district_crime_sf$rr_2020, district_crime
```

```
## Scale for fill is already present.
## Adding another scale for fill, which will replace the existing scale.

## Scale for fill is already present.
## Adding another scale for fill, which will replace the existing scale.
## Scale on map varies by more than 10%, scale bar may be inaccurate
##
## Scale on map varies by more than 10%, scale bar may be inaccurate
```



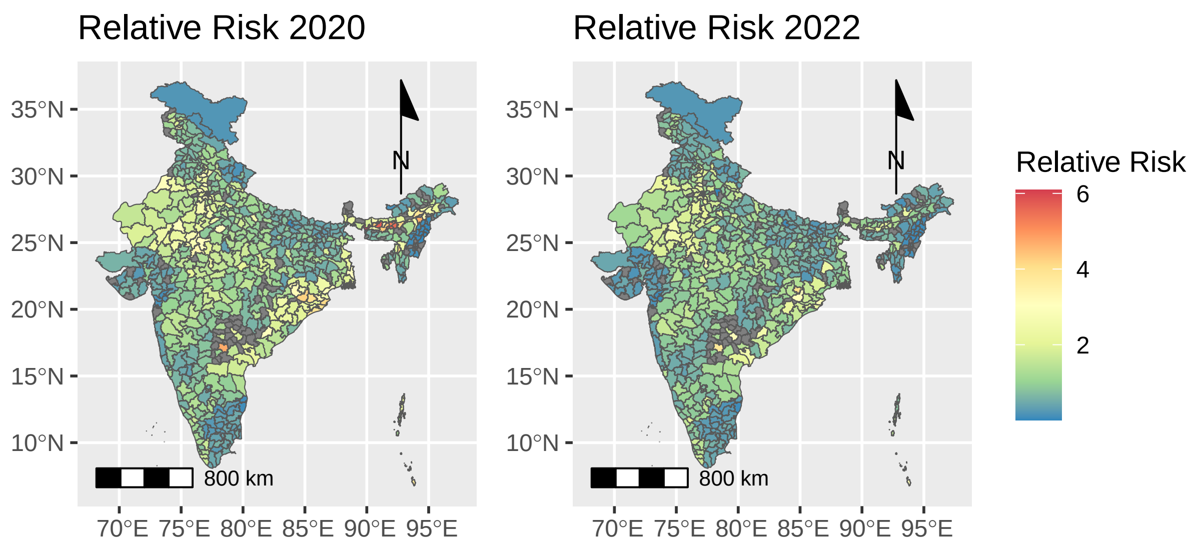

```r
# Final plot with title and common legend
rr_crime <- ((rr_2020 + theme(legend.position = "none")) + rr_2022) +
  plot_layout(ncol = 2,
              guides = "collect") +
  plot_annotation(title =
  "Prevalence Distribution of Relative Risk of crimes against women across districts and states in India
  caption = "Data Source: Data on crimes against women for the period 2020 and 2022 obtained from the Na
  theme = theme(plot.title = element_text(hjust = 0.125, size = 12))) &
  scale_fill_distiller("Relative Risk", palette = "Spectral", limits = range(c(district_crime_sf$rr_2020
```

```
## Scale for fill is already present.
## Adding another scale for fill, which will replace the existing scale.
## Scale for fill is already present.
## Adding another scale for fill, which will replace the existing scale.
```

```r
rr_crime
```

```
## Scale on map varies by more than 10%, scale bar may be inaccurate
## Scale on map varies by more than 10%, scale bar may be inaccurate
```

evalence Distribution of Relative Risk of crimes against women across districts and state



es against women for the period 2020 and 2022 obtained from the National Crime Records Bureau (NCRB) of India

```r
# Save the final plot
ggsave(
  filename = here("plots","rr_crime.png"),
  plot = last_plot(),
  scale = 1,
  width = 12,
  height = 9,
  dpi = 300,
)
```

```
## Scale on map varies by more than 10%, scale bar may be inaccurate
## Scale on map varies by more than 10%, scale bar may be inaccurate
```

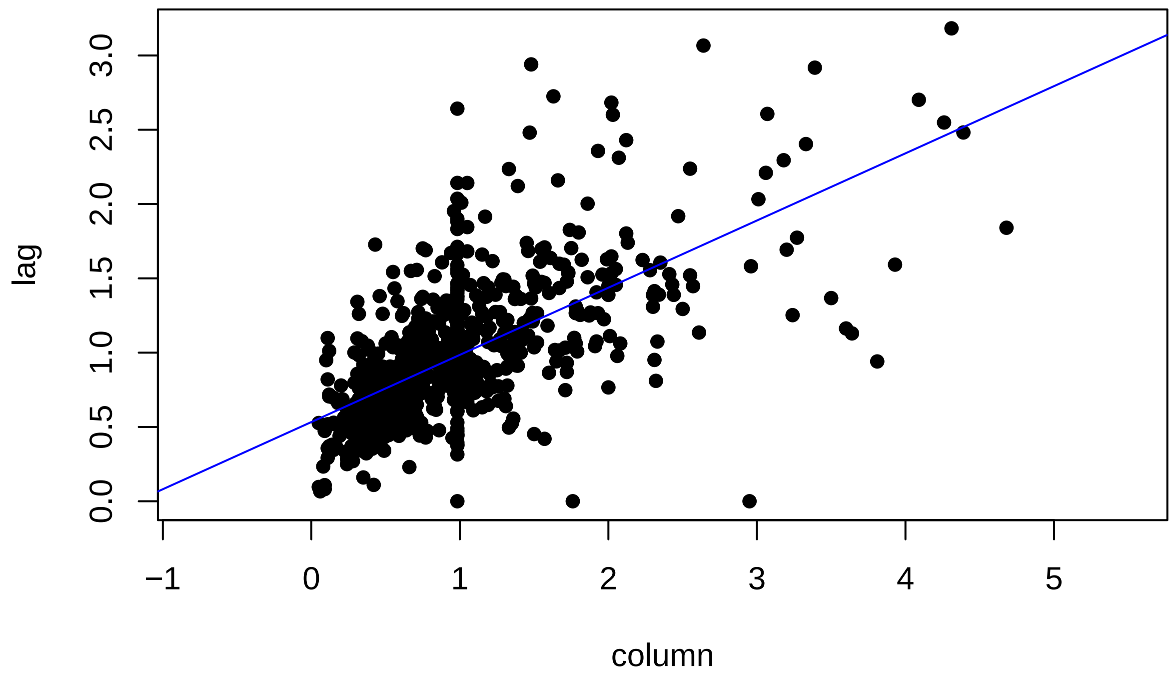
Moran's I

```r
moran_global <- function(data, column) {
  # Check for non-finite values (NA, NaN, Inf)
  non_finite_values <- !is.finite(column)

  # Impute with the mean (instance)
  column[!is.finite(column)] <- mean(column, na.rm = TRUE)

  # Replace Inf and -Inf with the maximum and minimum finite values
  column[column == Inf] <-
    max(column[is.finite(column)], na.rm = TRUE)
```

```r
  column[column == -Inf] <-
    min(column[is.finite(column)], na.rm = TRUE)

  # Define neighboring polygons # https://mgimond.github.io/simple_moransI_example/
  nb <- poly2nb(data, queen = TRUE)

  # Assign weights to the neighbors
  lw <- nb2listw(nb, style="W", zero.policy=TRUE)

  # Compute the (weighted) neighbor mean column values
  lag <- lag.listw(lw, column)


  # plot the relationship between column values and its spatially lagged counterpart.
  # The fitted blue line added to the plot is the result of an OLS regression model.

  plot(lag ~ column, pch = 16, asp = 1)

  M1 <- lm(lag ~ column)

  abline(M1, col = "blue")

  print("The slope of the line is the Moran’s I coefficient: ")
  print(coef(M1)[2])

  # Moran’s I analysis using the analytical method
  print(moran.test(column,lw, alternative="greater"))

}
```

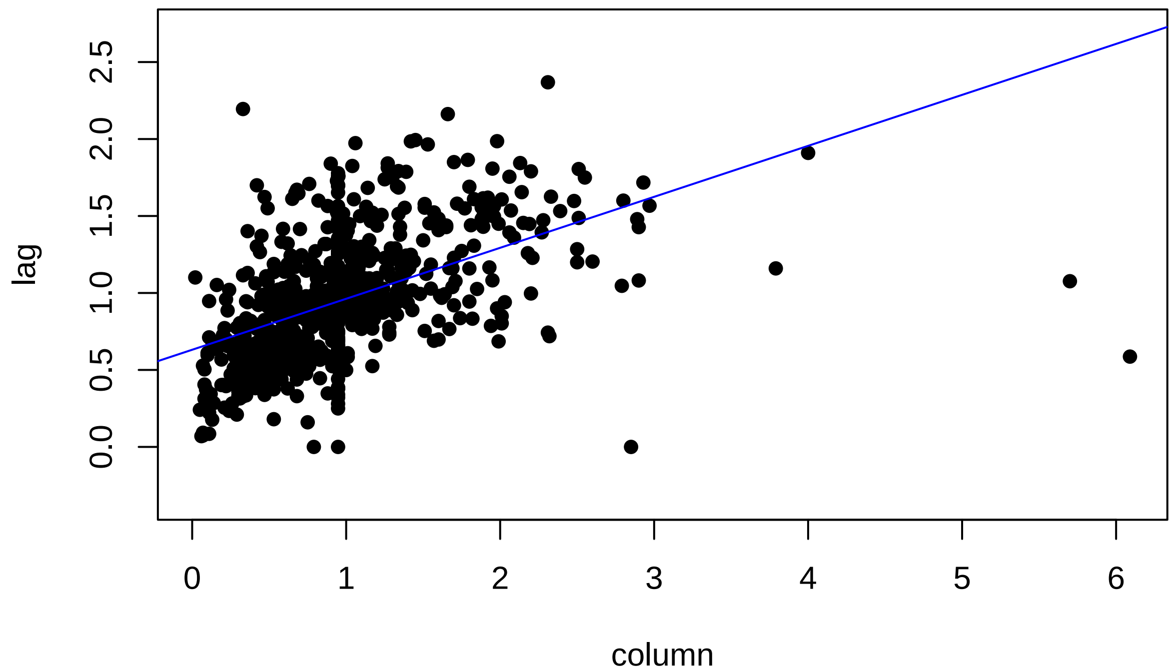
```r
# assign the variables
data   <- district_crime_sf
```

```r
# Calculate Global Moran
moran_global_rr_2020 <- moran_global(data, data$rr_2020)
```




```
## [1] "The slope of the line is the Moran's I coefficient: "
##    column
## 0.4520849
##
##  Moran I test under randomisation
##
## data:  column
## weights: lw
## n reduced by no-neighbour observations
##
## Moran I statistic standard deviate = 20.668, p-value < 2.2e-16
## alternative hypothesis: greater
## sample estimates:
## Moran I statistic       Expectation          Variance
##      0.4600622667     -0.0013831259      0.0004984862
```

```
moran_global_rr_2022 <- moran_global(data, data$rr_2022)
```




```
## [1] "The slope of the line is the Moran's I coefficient: "
##    column
## 0.3311331
##
##  Moran I test under randomisation
##
## data:  column
## weights: lw
## n reduced by no-neighbour observations
##
## Moran I statistic standard deviate = 15.221, p-value < 2.2e-16
## alternative hypothesis: greater
## sample estimates:
## Moran I statistic       Expectation          Variance
##      0.3369624672     -0.0013831259      0.0004941311
```

Local Moran Map

```
moran_local <- function(data, column) {
  # Check for non-finite values (NA, NaN, Inf)
  non_finite_values <- !is.finite(column)

  # Impute with the mean (instance)
  column[!is.finite(column)] <- mean(column, na.rm = TRUE)

  # Replace Inf and -Inf with the maximum and minimum finite values
```

```r
column[column == Inf] <-
  max(column[is.finite(column)], na.rm = TRUE)

column[column == -Inf] <-
  min(column[is.finite(column)], na.rm = TRUE)

# Define neighboring polygons # https://mgimond.github.io/simple_moransI_example/
nb <- poly2nb(data, queen = TRUE)

# Assign weights to the neighbors
lw <- nb2listw(nb, style = "W", zero.policy = TRUE)

# Compute the (weighted) neighbor mean column values
lag <- lag.listw(lw, column)

# Calculate Local Moran's I statistics for each spatial unit
local_moran <- localmoran(column, lw)

# Add Local Moran's I and p-values to the data set
data$local_moran_I <- local_moran[, 1]  # Local Moran's I values
data$local_moran_p <- local_moran[, 5]  # p-values

# Classify the Results https://spatialanalysis.github.io/handsonspatialdata/local-spatial-autocorrela

# Define significance level (e.g., 0.05)
significance_level <- 0.05

# Create a classification based on Local Moran's I and p-values
data$local_moran_category <- "Not Significant"

data$local_moran_category[data$local_moran_I > 0 &
                            data$local_moran_p <= significance_level] <- "High-High"

data$local_moran_category[data$local_moran_I < 0 &
                            data$local_moran_p <= significance_level] <- "Low-Low"

data$local_moran_category[data$local_moran_I > 0 &
                            data$local_moran_p > significance_level] <- "High-Low"

data$local_moran_category[data$local_moran_I < 0 &
                            data$local_moran_p > significance_level] <- "Low-High"

# Convert to factor for better visualization
data$local_moran_category <-
  factor(
    data$local_moran_category,
    levels = c(
      "Not Significant",
      "High-High",
      "Low-Low",
      "High-Low",
      "Low-High"
    )
```

```r
    )

  return(data)

}
```

```r
# Calculate Local Moran
moran_local_rr_2020 <- moran_local(data, data$rr_2020)
moran_local_rr_2022 <- moran_local(data, data$rr_2022)
```

```r
# LISA plot 2020
lisa_map_rr_2020 <- tm_shape(moran_local_rr_2020) +
  tm_polygons("local_moran_category",
              palette = c("#d9d9d9", "#e41a1c", "#377eb8", "#ffffb3", "#8dd3c7"),
              title = "LISA Map for Relative Risk 2020",
              style = "cat") +
  tm_borders() +
  tm_layout(legend.outside = TRUE,
            title = "Local Moran's I Clusters")  +
tm_scale_bar(position = c("left", "bottom"), width = "100 km", text.size = 0.5) +
tm_compass(position = c("right", "top"), type = "arrow", size = 2)

# LISA plot 2022
lisa_map_rr_2022 <- tm_shape(moran_local_rr_2022) +
  tm_polygons("local_moran_category",
              palette = c("#d9d9d9", "#e41a1c", "#377eb8", "#ffffb3", "#8dd3c7"),
              title = "LISA Map for Relative Risk 2022",
              style = "cat") +
  tm_borders() +
  tm_layout(legend.outside = TRUE,
            title = "Local Moran's I Clusters")  +
tm_scale_bar(position = c("left", "bottom"), width = "100 km", text.size = 0.5) +
tm_compass(position = c("right", "top"), type = "arrow", size = 2)
```

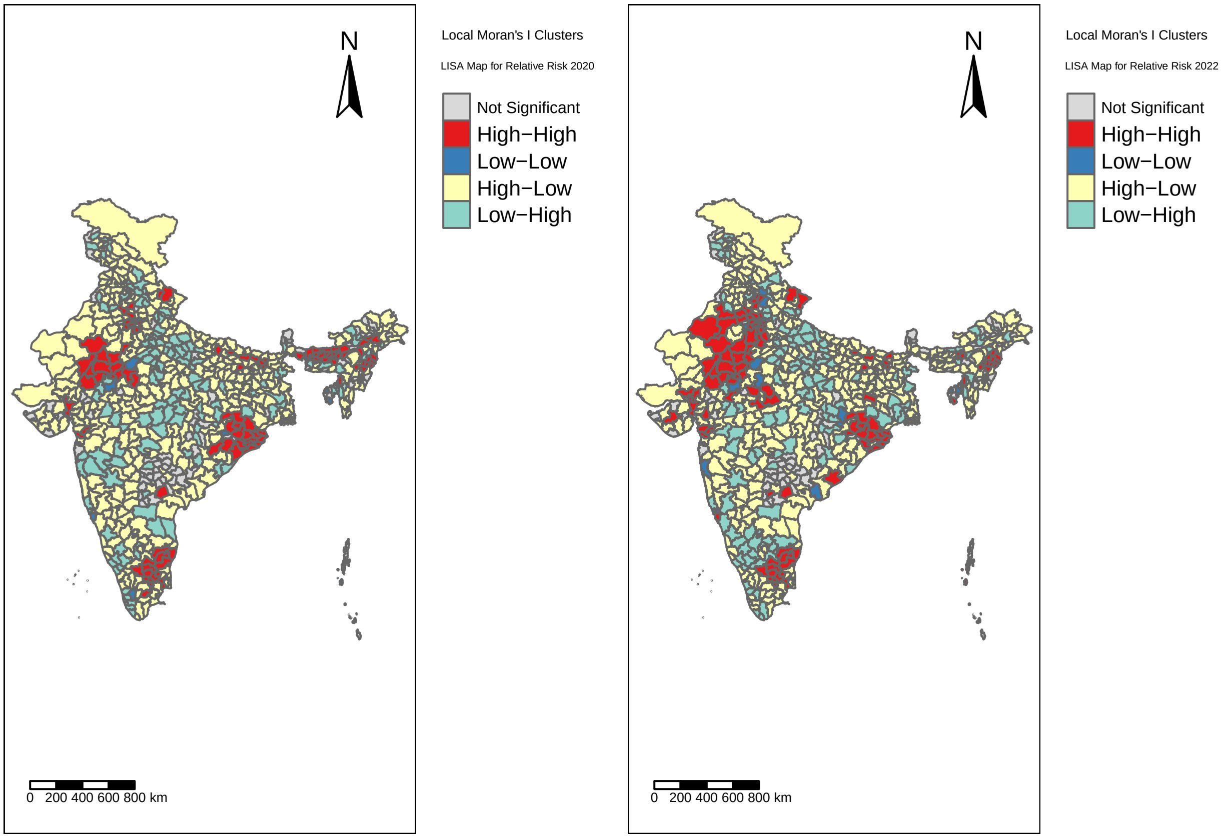
```r
# Combine the plots
combined_lisa_map <- tmap_arrange(
  lisa_map_rr_2020,
  lisa_map_rr_2022,
  ncol = 2,
  # Arrange the maps in 2 columns (side by side)
  widths = 24,
  heights = 9,
  sync = FALSE,
  asp = 0,
  outer.margins = 0.02
)

combined_lisa_map
```

```
## Warning: One tm layer group has duplicated layer types, which are omitted. To
## draw multiple layers of the same type, use multiple layer groups (i.e. specify
## tm_shape prior to each of them).
```

```
## Scale bar width set to 0.25 of the map width
```

```
## Scale bar set for latitude km and will be different at the top and bottom of the map.
```

```
## Some legend labels were too wide. These labels have been resized to 0.53. Increase legend.width (argu

## Warning: One tm layer group has duplicated layer types, which are omitted. To
## draw multiple layers of the same type, use multiple layer groups (i.e. specify
## tm_shape prior to each of them).

## Scale bar width set to 0.25 of the map width

## Scale bar set for latitude km and will be different at the top and bottom of the map.

## Some legend labels were too wide. These labels have been resized to 0.53. Increase legend.width (argu
```




```r
?tm_layout
```

```
## starting httpd help server ... done
```

```r
# Save the plot
tmap_save(combined_lisa_map, filename = here("plots","rr_combined_lisa_map.png"), scale = 1, width = 24
```

```
## The argument 'units' has been set to "in" since the specified width or height is less than or equal t

## Warning: One tm layer group has duplicated layer types, which are omitted. To
## draw multiple layers of the same type, use multiple layer groups (i.e. specify
## tm_shape prior to each of them).

## Scale bar width set to 0.25 of the map width

## Scale bar set for latitude km and will be different at the top and bottom of the map.

## Warning: One tm layer group has duplicated layer types, which are omitted. To
## draw multiple layers of the same type, use multiple layer groups (i.e. specify
## tm_shape prior to each of them).
```

```
## Scale bar width set to 0.25 of the map width
## Scale bar set for latitude km and will be different at the top and bottom of the map.
## Map saved to D:\R Urban Analytics\SOM 787\Excercises\plots\rr_combined_lisa_map.png
## Resolution: 7200 by 2700 pixels
## Size: 24 by 9 inches (300 dpi)
```

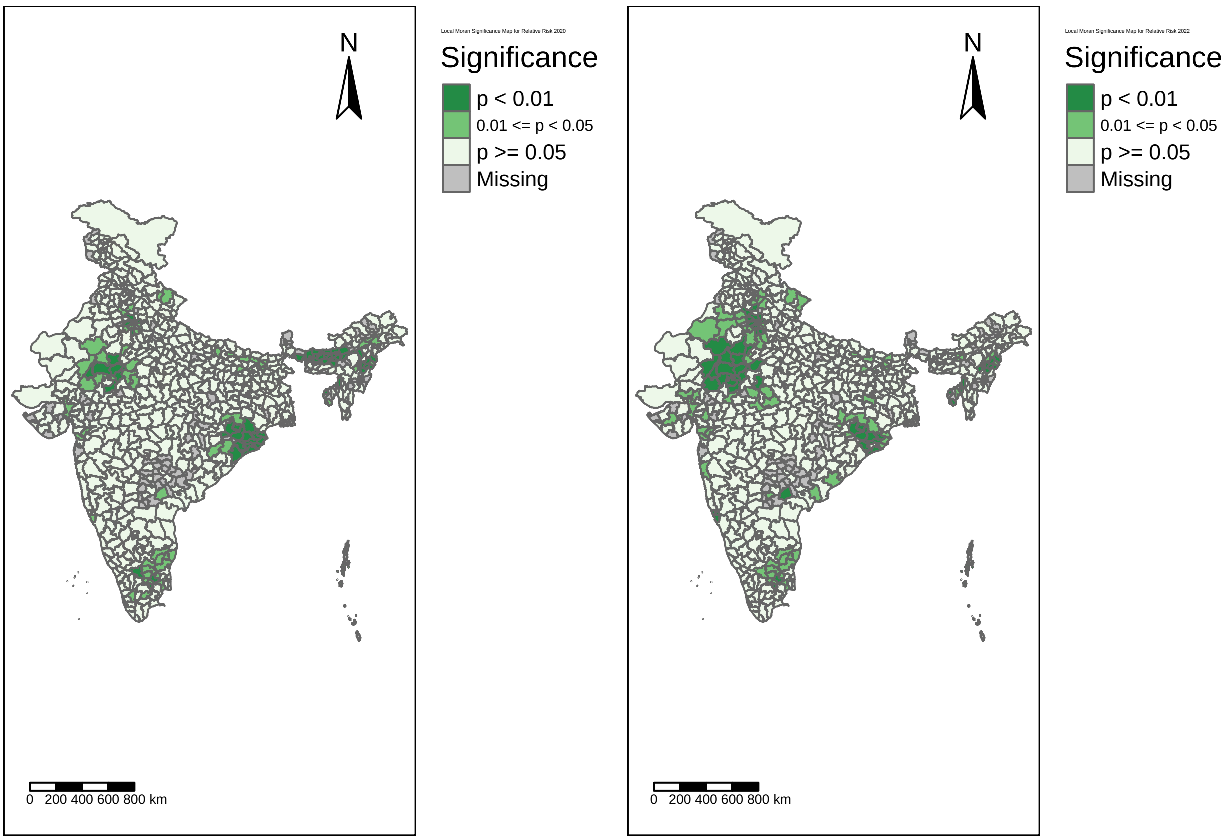
Local Moran Map

```
# Create the Significance Map

# 2020
significance_map_rr_2020 <- tm_shape(moran_local_rr_2020) +
  tm_polygons("local_moran_p",
              palette = c("#238b45", "#74c476", "#edf8e9"),
              breaks = c(0, 0.01, 0.05, 1),
              labels = c("p < 0.01", "0.01 <= p < 0.05", "p >= 0.05"),
              title = "Significance") +
  tm_borders() +
  tm_layout(legend.outside = TRUE,
            title = "Local Moran Significance Map for Relative Risk 2020")  +
tm_scale_bar(position = c("left", "bottom"), width = "100 km", text.size = 0.5) +
tm_compass(position = c("right", "top"), type = "arrow", size = 2)


# 2022
significance_map_rr_2022 <- tm_shape(moran_local_rr_2022) +
  tm_polygons("local_moran_p",
              palette = c("#238b45", "#74c476", "#edf8e9"),
              breaks = c(0, 0.01, 0.05, 1),
              labels = c("p < 0.01", "0.01 <= p < 0.05", "p >= 0.05"),
              title = "Significance") +
  tm_borders() +
  tm_layout(legend.outside = TRUE,
            title = "Local Moran Significance Map for Relative Risk 2022")  +
tm_scale_bar(position = c("left", "bottom"), width = "100 km", text.size = 0.5) +
tm_compass(position = c("right", "top"), type = "arrow", size = 2)
```

```
# Combine the plots
combined_significance_map <- tmap_arrange(
  significance_map_rr_2020,
  significance_map_rr_2022,
  ncol = 2,
  # Arrange the maps in 2 columns (side by side)
  widths = 24,
  heights = 9,
  sync = FALSE,
  asp = 0,
  outer.margins = 0.02
)

combined_significance_map
```

```
## Warning: One tm layer group has duplicated layer types, which are omitted. To
```

```
## draw multiple layers of the same type, use multiple layer groups (i.e. specify
## tm_shape prior to each of them).

## Scale bar width set to 0.25 of the map width

## Scale bar set for latitude km and will be different at the top and bottom of the map.

## Some legend labels were too wide. These labels have been resized to 0.47. Increase legend.width (argu

## Warning: One tm layer group has duplicated layer types, which are omitted. To
## draw multiple layers of the same type, use multiple layer groups (i.e. specify
## tm_shape prior to each of them).

## Scale bar width set to 0.25 of the map width

## Scale bar set for latitude km and will be different at the top and bottom of the map.

## Some legend labels were too wide. These labels have been resized to 0.47. Increase legend.width (argu
```




```r
# Save the plot
tmap_save(combined_significance_map, filename = here("plots","rr_combined_significance_map.png"), scale
```

```
## The argument 'units' has been set to "in" since the specified width or height is less than or equal t

## Warning: One tm layer group has duplicated layer types, which are omitted. To
## draw multiple layers of the same type, use multiple layer groups (i.e. specify
## tm_shape prior to each of them).

## Scale bar width set to 0.25 of the map width

## Scale bar set for latitude km and will be different at the top and bottom of the map.
```

```
## Warning: One tm layer group has duplicated layer types, which are omitted. To
## draw multiple layers of the same type, use multiple layer groups (i.e. specify
## tm_shape prior to each of them).

## Scale bar width set to 0.25 of the map width
## Scale bar set for latitude km and will be different at the top and bottom of the map.

## Map saved to D:\R Urban Analytics\SOM 787\Excercises\plots\rr_combined_significance_map.png

## Resolution: 7200 by 2700 pixels

## Size: 24 by 9 inches (300 dpi)
```